R1

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1 New Code / Devs

1.1 Python Terminal Script for Filtering

Due to the fact that we to filter multiple simulations with very similar methods, I decided to make an actual script instead of just using a notebook repeatably. The largest upside is that this can easily be run on a compute node, which makes filtering larger sims easier. Also helps make sure any results are reproducable.

Takes a user input via the CLI (see below), filters for all `BH_Sol` systems, and returns the population file with said systems. Additionally, it returns a .csv of just the `BH_Sol` rows. While running it provides feedback via the terminal stating how many `BH_MS` systems it found, as well as how many are the expected S1:S2 configuration. If any are S2:S1, it prints a warning as well as the total count.

Need to further refine it, but there's also a default SBATCH script to help expidite the submission process.

Usage Example

```
python BH-Sol_Filter.py /Data/1e+00_Zsun_population.h5 -o ./testResults
```

Arguments

Argument	Use	Ex value
`-o`	output location	.results
`-outputName`	name for the outputed files[1]	`BH_Sol_Systems`
`-overwrite`	Whether to overwrite existing results, T/F bool. Defaults to F	True
`--maxMass`	Max mass used when filtering, defaults to $3M_\odot$	3.5
`--maxPeriod`	Max period used when filtering, defaults to 3.5 days	4

Can be found at https://github.com/PiersonLip/stellarBHs/blob/main/FilteringScripts/BH-Sol_Filter.py

[1] If a name is not specificed it will use the orignal filename, appending a specifier to the front

2 New Research

2.1 Pop 1

When filtering via notebook and script, both returned 6 systems that had $p_{orb} < 3.5, M_{\odot,2} < 3, t < 10e9$

These systems all evolved via the same channel, `ZAMS_oCE1_CC1_oRLO2_CC2`. Via inspection of the history df and the oneline, it is evident that the systems evolve detached, then into post-MS **unstable** RLO via S1 $\rightarrow$ S2. This period decrease leads to common envelope, which then *greatly* decreases the orbtial period. The envelope is then ejected, leaving a depleted stripped He core[2]. The core then collapses in a couple of thousand years, leaving the system in a BH-MS state with low p_{orb}/seperation.

These systems then evolve in a semi-peculiar way, where it is highy likely that there will be RLO from S2 onto S1, and because $S1 > S2$, the transfer is stable, leading to *very* prolonged stable RLO (seemingly on the scale of $> 10Gyr$), which leaves S2 at low masses ($< .08M_{\odot}$).

Interestingly, the inital period for all the systems is close to $4000d$, with mean of 4022 and a std of only 151. However, a larger sample size is needed to confirm this.

Additionally, the inital masses of S1 and S2 appear to be quite constrained as well.

	orbital_period_i
count	6.000000
mean	4022.716197
std	151.522384
min	3873.118288
25%	3908.300983
50%	3978.305326
75%	4124.397101
max	4248.507680

	S1_mass_i
count	6.000000
mean	17.515761
std	0.520269
min	16.815111
25%	17.190036
50%	17.510176
75%	17.818155
max	18.254965

	S2_mass_i
count	6.000000
mean	2.296788
std	0.320702
min	1.797809
25%	2.120753
50%	2.353828
75%	2.547210
max	2.620529

2.2 Pop 2

Ran a second grid inlcuding an eccentiricity distro (Ps and Qs), as well as slightly constraining the dataset further ($12 < M_1 < 100, \quad .1 < M_2 < 10$)[3], hoping to confirm some of the init conditions.

	orbital_period_i
count	12.000000
mean	4136.066721
std	177.676729
min	3844.570082
25%	4024.214908
50%	4119.397535
75%	4273.345845
max	4370.857404

	S1_mass_i
count	12.000000
mean	2.240382
std	0.412075
min	1.551266
25%	1.903135
50%	2.300024
75%	2.579111
max	2.729828

	S2_mass_i
count	12.000000
mean	17.357478
std	1.073496
min	15.506937
25%	16.714083
50%	16.998229
75%	18.162633
max	19.435080

[2] I need to read a lot more about CE evo, but my intutuion feels weird about the fact that just the envolope of *one* of the stars is ejected

[3] This is very constrained very early, but I wanted to try something while waiting for the 10 million sim

2.3 Full P_{orb}

When filterting pop 2 with a max $p_{orb} = 300$days we find 46 BH-Sol systems, all of these (except one!) evolve via the same CE channel. There is one system which does *not* do RLO before CC1, and it also only reaches a solar-like mass after RLO2, so I'm exlucding it for now. Probably important to keep in mind though.

When plotting S1 vs S2 intial and formation masses there appears be a correlation. In fig 2.3 this correlation appears to be quite strong, however, this very well may be becuase of the intial distribution.

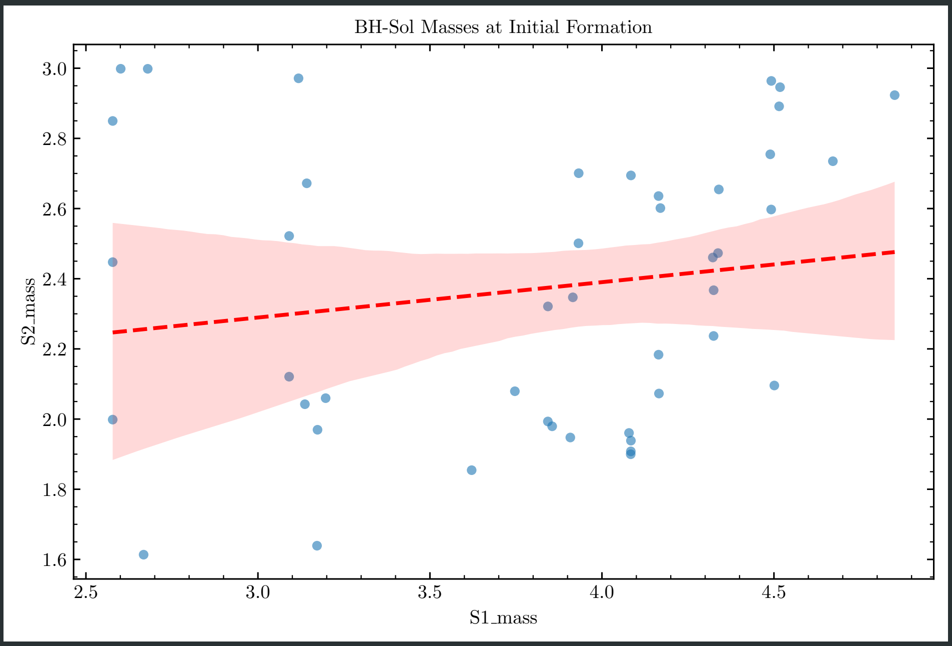

Figure 1: Masses at intial BH-Sol state. i.e. BH formation/S1 collaspe

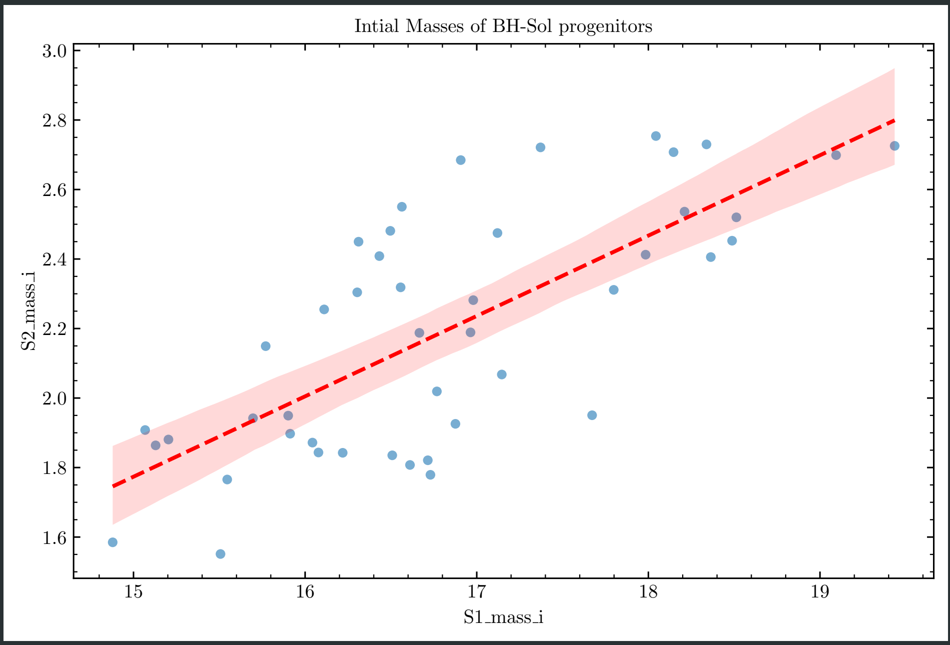

Figure 2: Initial ZAMS masses. May just be a byproduct of Ps and Qs, further sims with diff intial distribution needed.

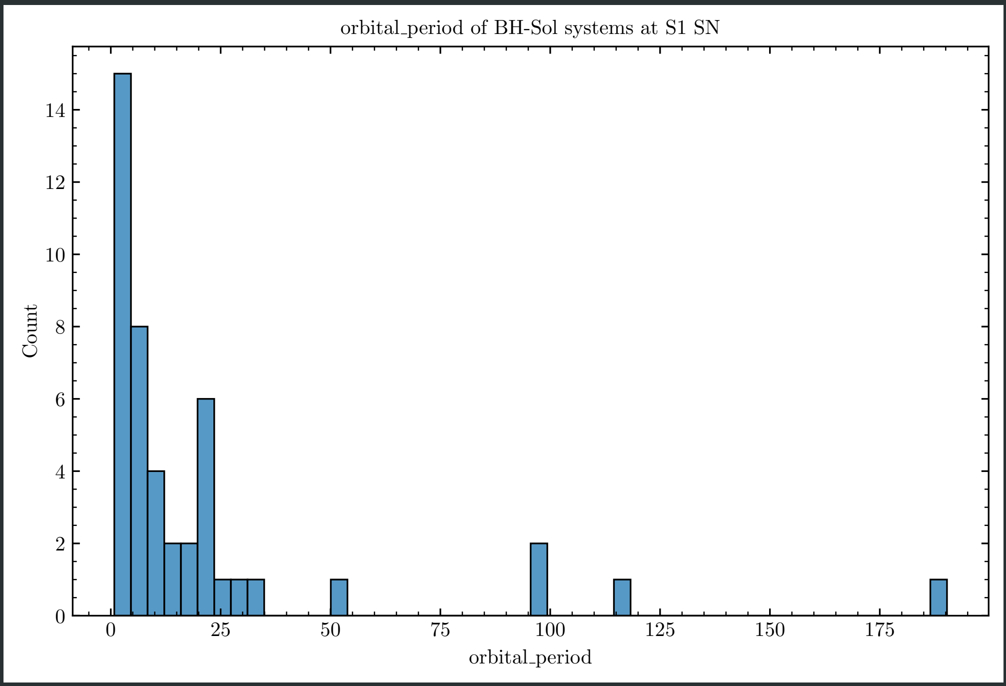

Figure 3: Irbital Period of BH-Sol Systems at init onsent.

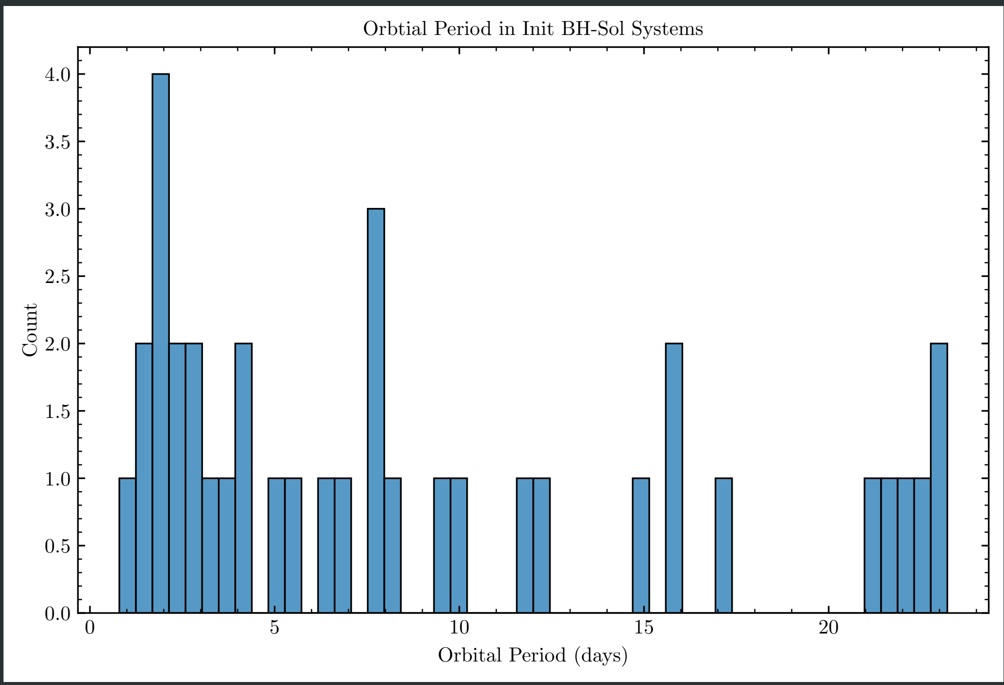

Figure 4: Crop of orbital Period of BH-Sol Systems at init onsent.

3 To-Do

- Fid out if it makes sense for the onset of un-stable rlo to happen at $P_{orb} \approx 4000$days

4 Questions

- Tried running a grid, seems to have just completly failed from trying to refence itself before it was fully initalized. Entire log was filled with

```
    Traceback (most recent call last):
    File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/popsyn/binarypopulation.py", 
      binary.evolve()
    File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", l
      self.run_step()
    File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", l
      next_step(self)
    File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_merged.py
      super().__call__(binary)
    File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_isolated.
      super().__call__(binary)
    File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.
      self.update_after_evo(t, binary, primary, secondary)
    File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.
      current = history[-1]
                ~~~~~~~~~~~
  IndexError: index -1 is out of bounds for axis 0 with size 0
```

  and from what I can tell, this happened to nearly all the simmed bianaries

- Should we include an inital eccentiricity distribution? I imagine it would impact formation with unstable RLO and CE.

- Does it make sense to still be filtering for $p_{orb} < 3$days? The simulations seem to be suggesting that *all* BH-Sol systems evolve through CE, and that the main thing effecting p_{orb} post CE is the kick strength of the BH. So, maybe we should be initally filterting for all BH-Sol, and if we want the formation rate of specifically low p_{orb} ones, just take the subpop.

- Cool if i present an assortment of general POSYDON things ive worked on at astrofest?